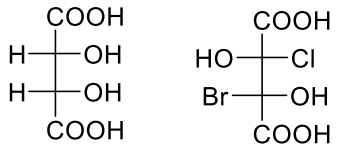
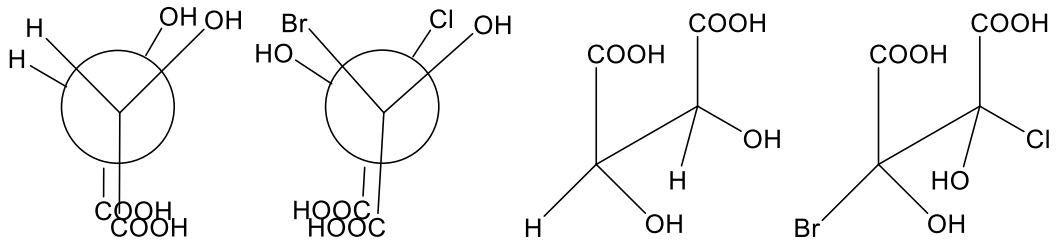
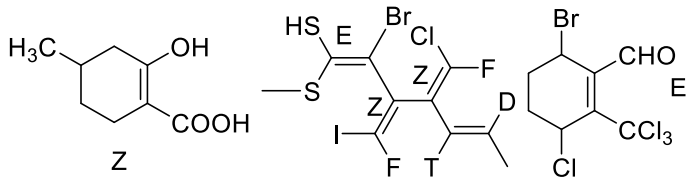
Tutorial 1 (CH1201)

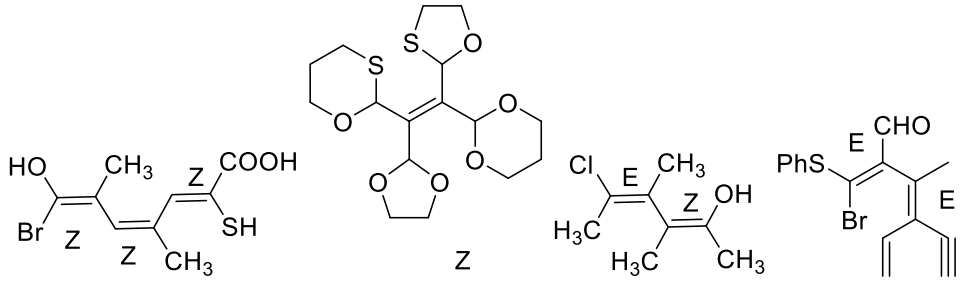
1. Convert the following Fischer projections to Newman and Saw-horse projection.



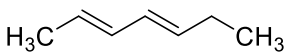


2. Denote the configuration (E/Z) of following molecules with the support of CIP rules.

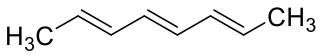




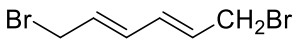
3. Calculate the number of geometrical isomers of the given molecules.



a) **No of GI = 4**



b) **No of GI = 6**
Symmetrical structure
Formula $2^{n-1}+2^{n-1/2}$



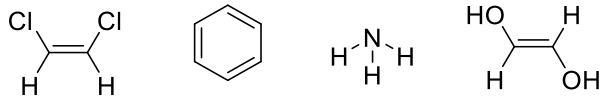
c) No. of GI = 3
Symmetrical structure
Even no.of double bonds
Formula $2^{n-1}+2^{n/2-1}$

a is unsymmetrical structure, the formula is 2^n (n = no.of double bonds)

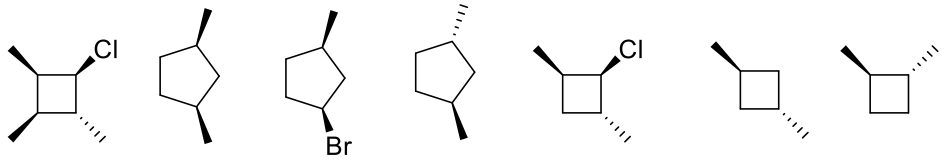
b is symmetrical structure with odd number of double bonds

c is symmetrical structure with even number of double bonds

4. Identify the symmetry elements present in the following molecules.



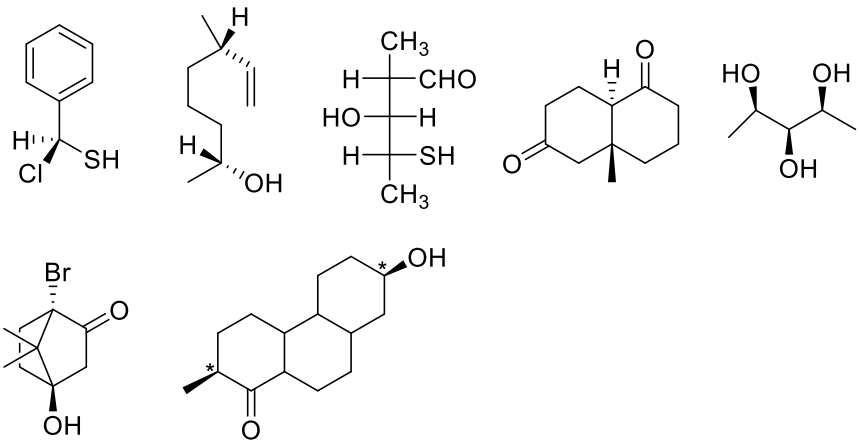
5. Which of the following is optically active.



6. Name five capital letters which exhibit enantiomer relation.

Tutorial 3 (CH1201)

1. Predict the R/S-configuration of the following compounds.



2. Estimate the ee% and total % of R isomer using following data; specific rotation of R isomer is -15.2°, observed rotation of the sample is -2.8°.

3. (+)-Mandaleic acid has a specific rotation of +158°. What would be the observed specific rotation of each of the following mixtures:

a) 25% (-)-Mandelic acid and 75% (+)-Mandelic acid

b) 50% (-)-Mandelic acid and 50% (+)-Mandelic acid

c) 75% (-)-Mandelic acid and 25% (+)-Mandelic acid

4. A solution prepared by mixing 10 mL of 0.10 M solution of the R-enantiomer and 30 mL of a 0.10 M solution of the S-enantiomer was found to have an observed specific rotation of +4.8°. What is the specific rotation of each of the enantiomer?

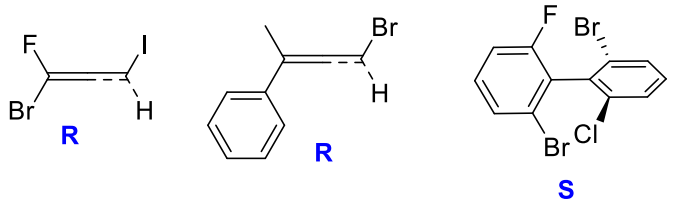
5. The specific rotation of R-(+)-Lactic acid is +8.7o. If the observed specific rotation of a mixture of R-Lactic acid and S-Lactic acid is +1.4o, what percent of Lactic acid present as the R-enantiomer?

Tutorial 2 (CH1101)

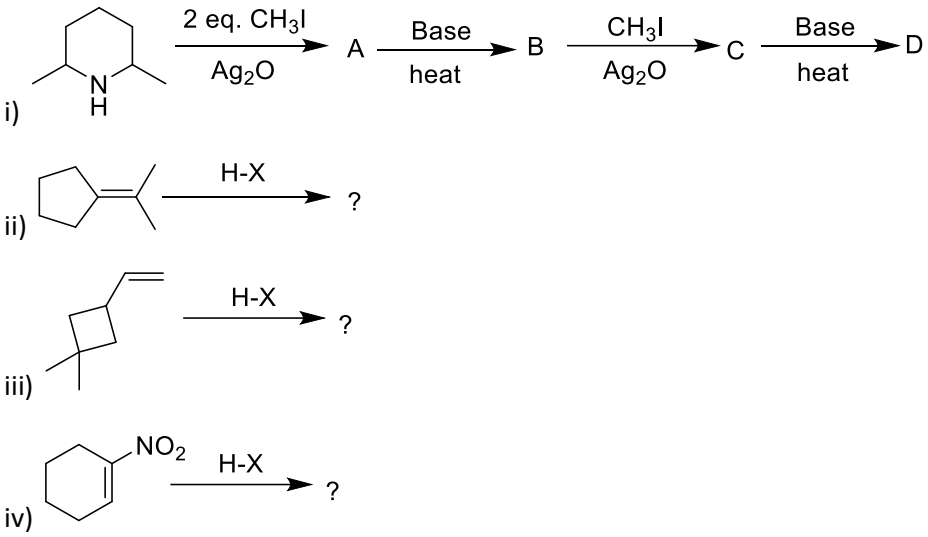
1. Explain the confirmational analysis of n-Butane with energy level diagram.

2. Explain the confirmational analysis of 1,3-dimenthlcyclohexane

3. Predict the configuration for following



4. Complete the following reactions with detailed mechanism.

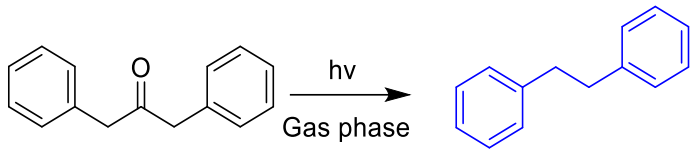


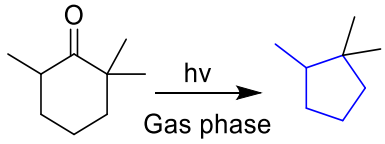
Tutorial 4 (CH1201)

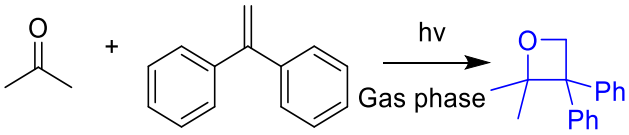
1. Sketch the Jablonski diagram and explain the electronic transitions in it.

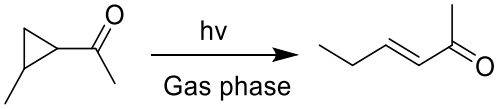
2. Define Grotthurs-Drapper and Einstein Photochemical Equivalence laws.

3. Complete the following reactions and explain their mechanism.

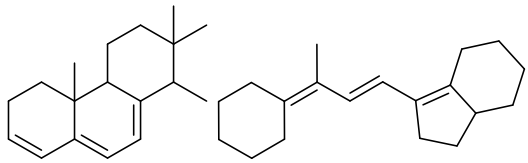








4. Calculate the λ_{max} of the following compounds.



Ref values: Homoannular = 253 nm, Heteroannular = 214 nm, alkyl substitution = 5 nm, Exo-double bond = 5 nm, Butadiene = 217 nm, double bond extension = +30 nm